

Adjugate of a Matrix

Let A be an $n \times n$ matrix. Let $C = (c_{ij})$ be the matrix such that

$$c_{ij} = (-1)^{i+j} \det(A_{ij}).$$

Then C^T is called the adjugate of A .

The number c_{ij} is called the ij^{th} Cofactor of A .

The matrix C is called the Cofactor matrix of A .

Theorem: Let A be an $n \times n$ matrix and C be the Cofactor matrix of A . Then $AC^T = C^T A = \det(A) I_n$.

Proof Let $B = AC^T$ and denote the ij entry of B by b_{ij} . Then

$$\textcircled{A} \leftarrow b_{ij} = \sum_{k=1}^n a_{ik} \cdot c_{jk} = \sum_{k=1}^n a_{ik} (-1)^{j+k} \det(A_{jk})$$

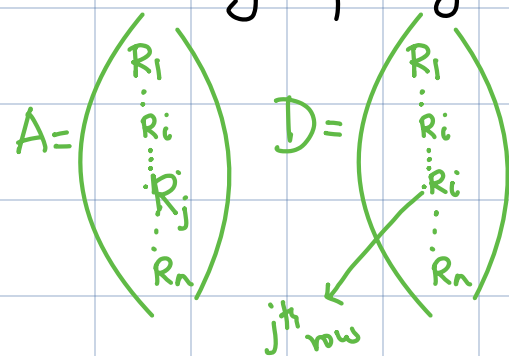
What we need to show is that $B = \det(A) I_n$. That is,

$$b_{11} = b_{22} = \dots = b_{nn} = \det(A) \text{ and } b_{ij} = 0 \text{ if } i \neq j.$$

* When $i=j$, we have

$$b_{ii} = \sum_{k=1}^n (-1)^{i+k} a_{ik} \det(A_{ik}) = \det(A). \quad \checkmark$$

* When $i \neq j$, we consider the matrix D obtained by replacing j^{th} row of A by i^{th} row of A .



Then $\det(D) = 0$. Also,

$$\begin{aligned} 0 = \det(D) &= \sum_{l=1}^n (-1)^{j+l} d_{jl} \det(D_{jl}) \\ &= \sum_{l=1}^n (-1)^{j+l} d_{jl} \det(A_{jl}) = \sum_{l=1}^n (-1)^{j+l} a_{il} \det(A_{jl}) \end{aligned}$$

Thus, from (A), we obtain $b_{ij} = 0$ for $i \neq j$ $\square$

Corollaries

① If A is invertible then

$$A^{-1} = C^T \cdot \frac{1}{\det(A)}$$

② You can expand the determinant via any column as well (use transpose).

HW Project. Learn about Cramer's rule and write a report on it not exceeding 3 pages.